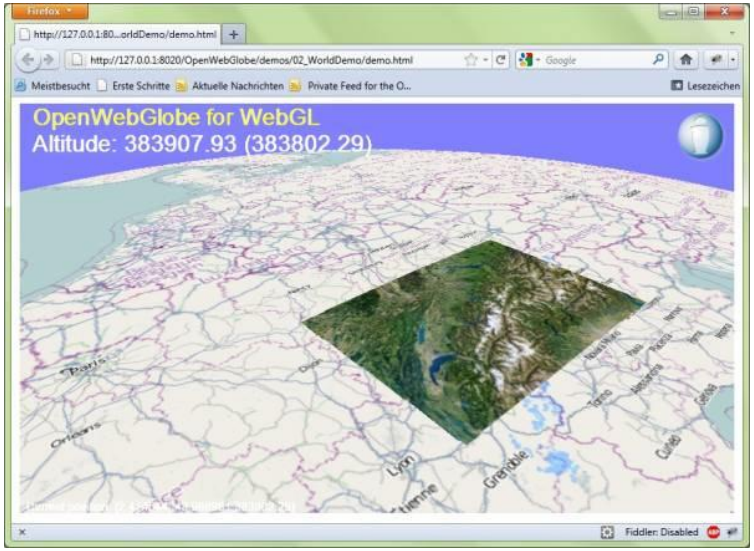
University of Applied Sciences Northwestern Switzerland
School of Architecture, Civil Engineering and Geomatics

OpenWebGlobe SDK
An Open Source High-Performance Virtual Globe SDK for Open Maps





Martin Christen
University of Applied Sciences Northwestern Switzerland
martin.christen@fhnw.ch

© 2011 FHNW / IVGI

Introduction: About OpenWebGlobe – Open Source Virtual Globe





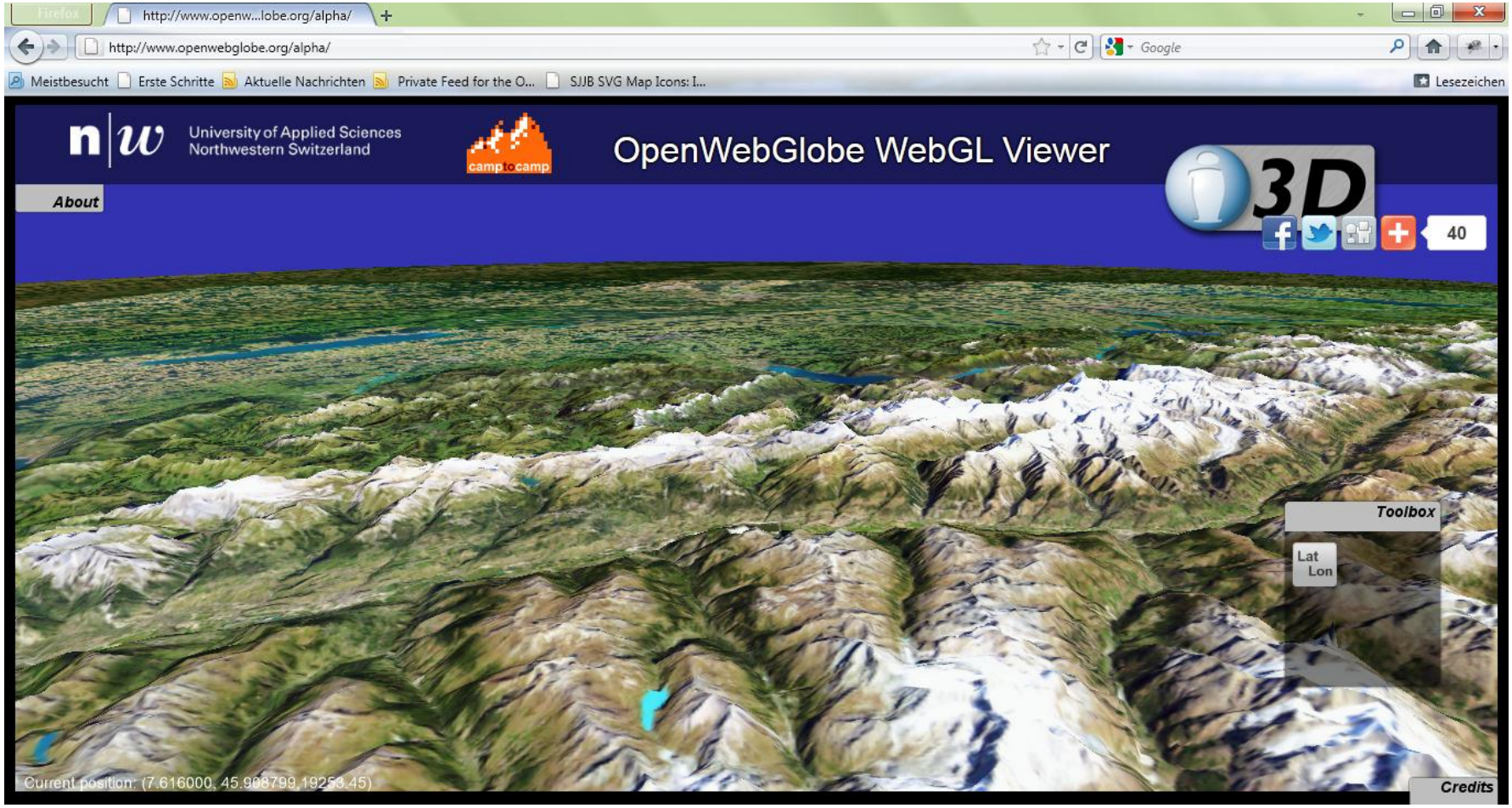




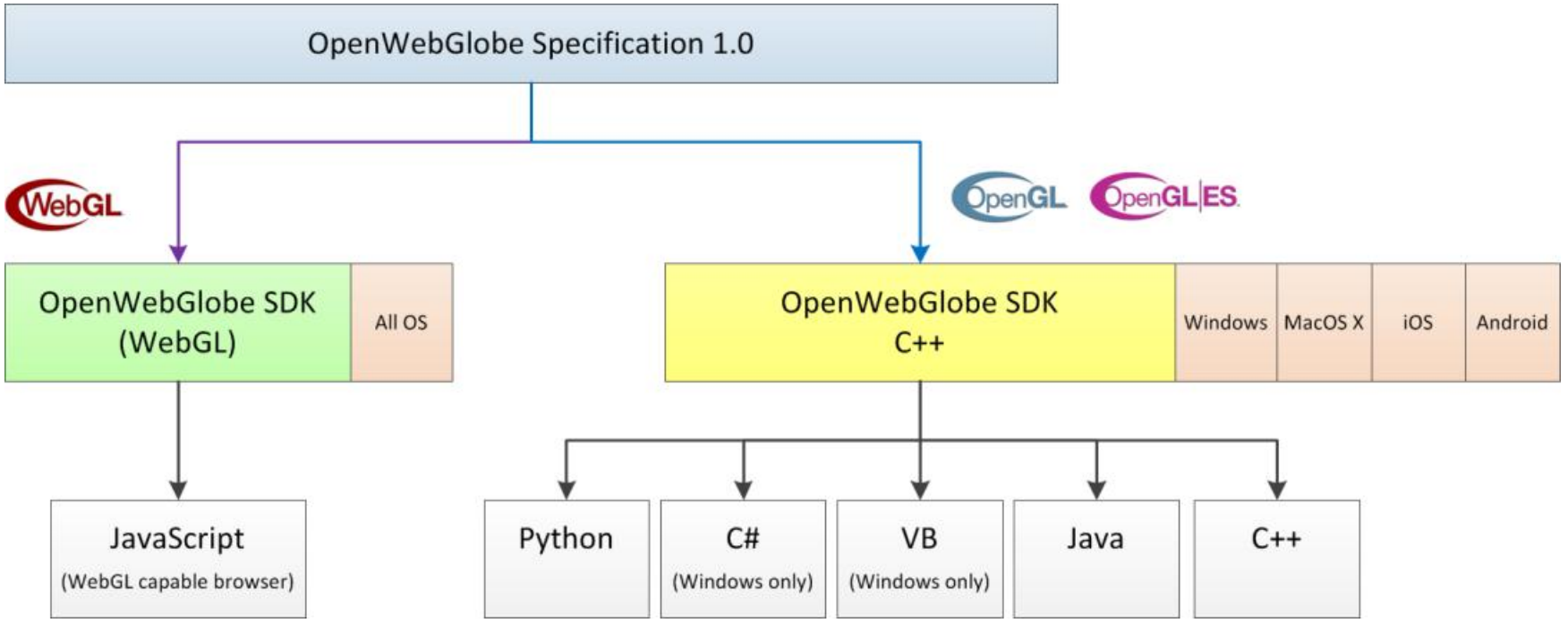
Videos on:
http://www.openwebglobe.org

OpenWebGlobe in the Web Browser

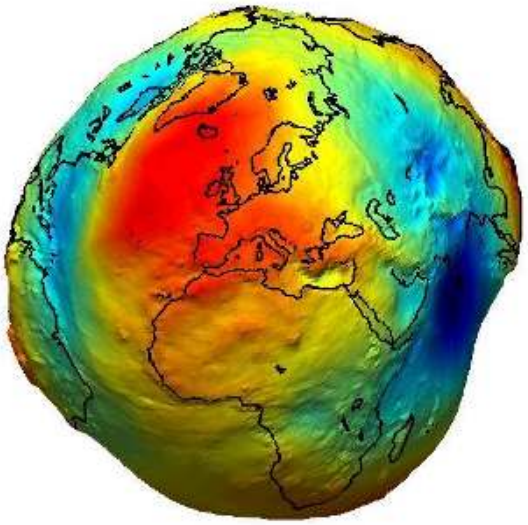
http://www.openwebglobe.org/alpha

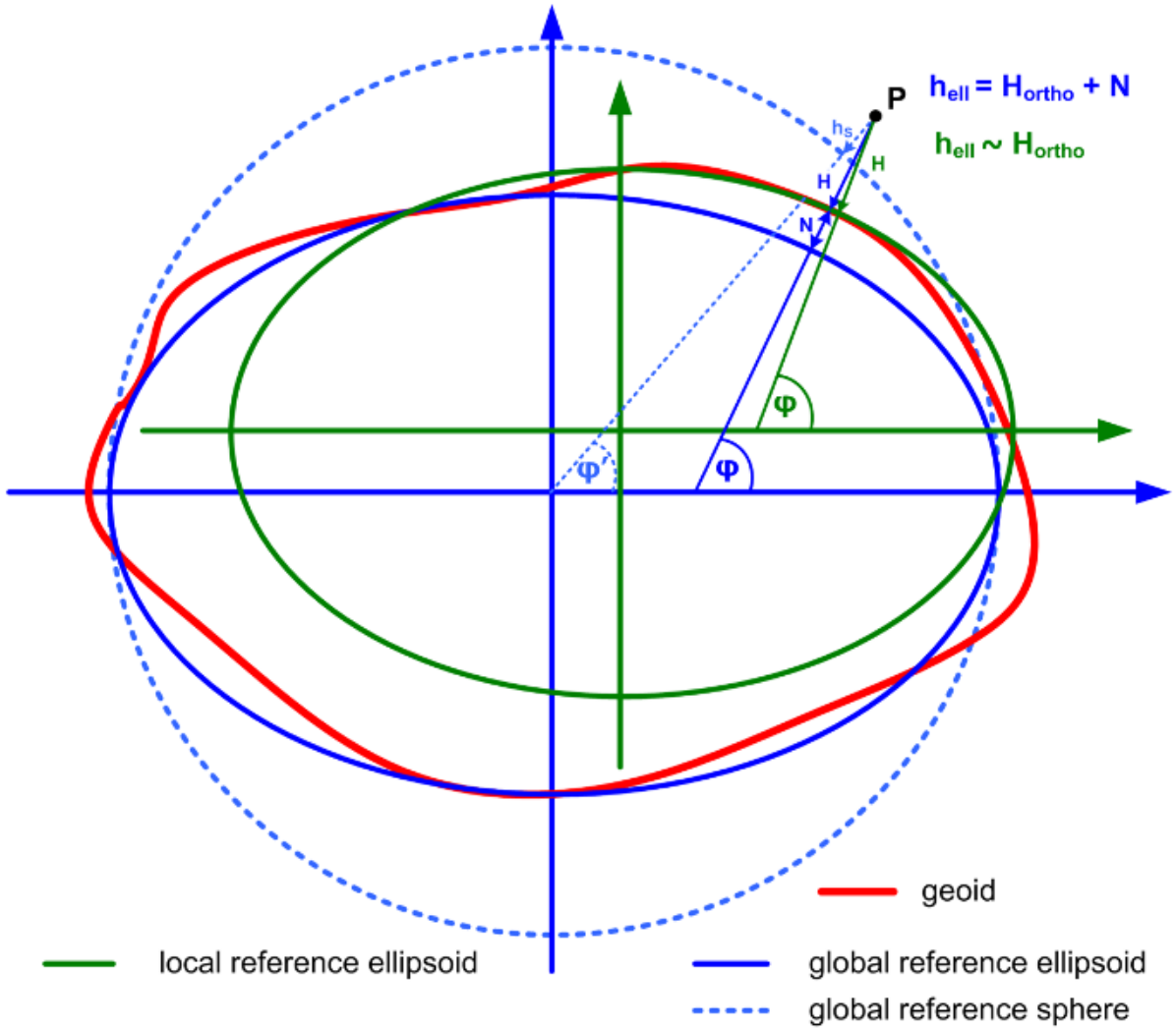


OpenWebGlobe Technology



OpenWebGlobe – Geodetic Reference Systems





- **Geodetic Reference Systems in Virtual Globes**

- **Model in i3D**

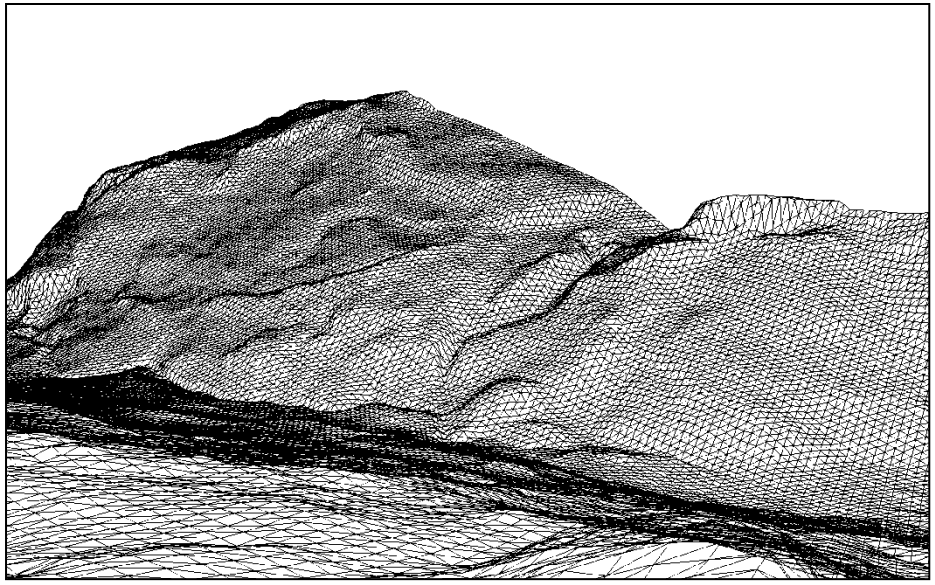
- WGS-84 ellipsoidal model &

- global geoid model

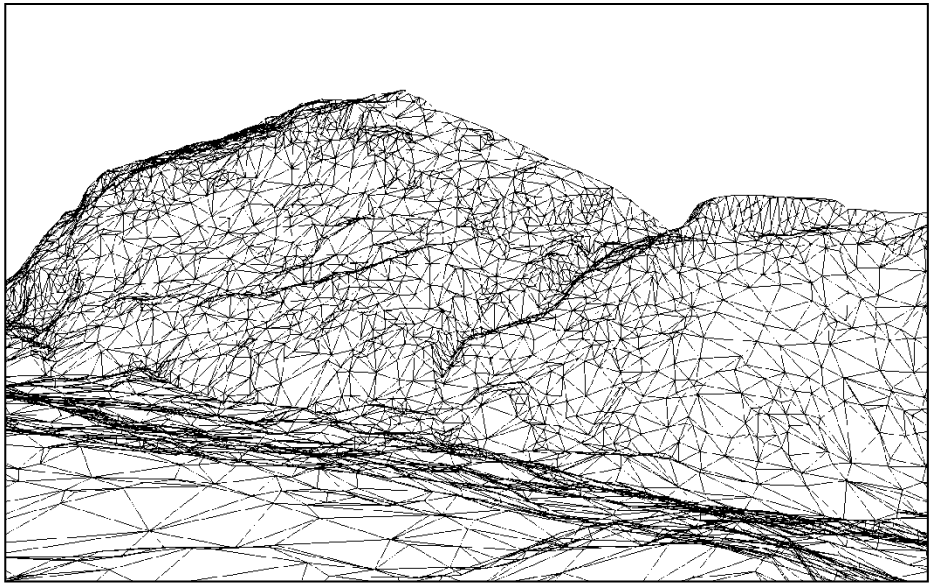
for high accuracies position **and** height

Elevation Data Processing

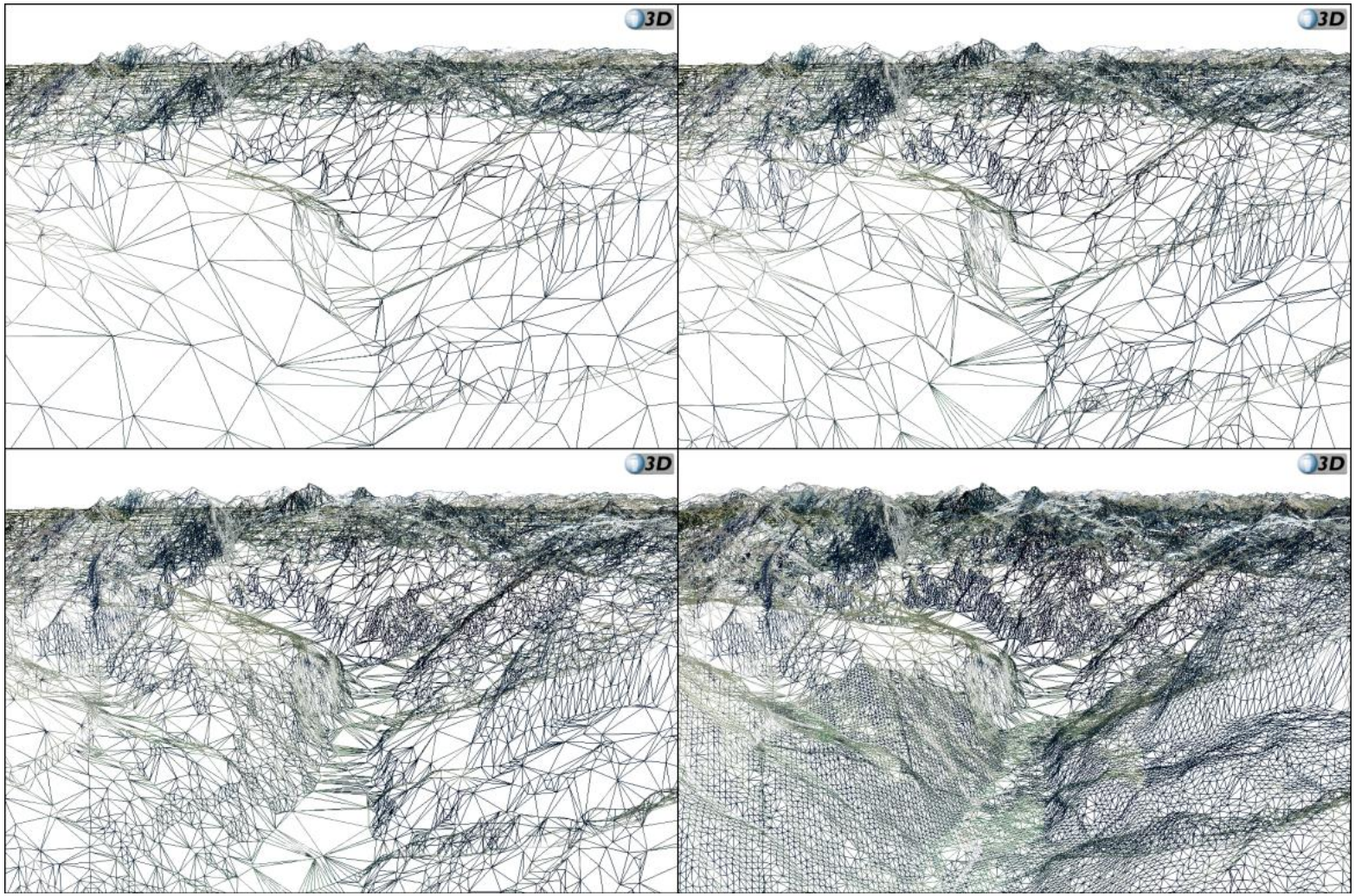
GRID



TIN



Elevation: Interactive Quality - from mobile devices to high end computers



A Simple “Hello World” using the OpenWebGlobe SDK

```
context = ogCreateRenderWindow("Hello Globe", 640, 480);

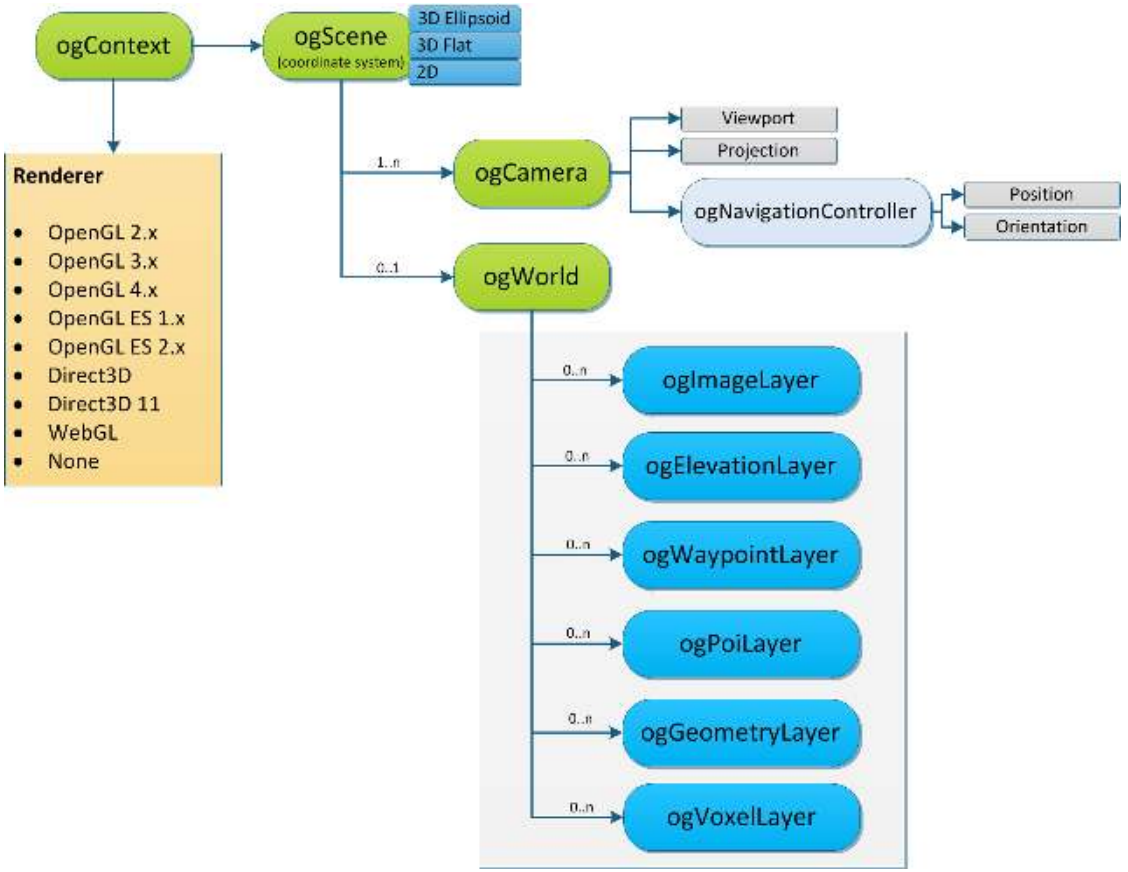
globe = ogCreateGlobe(context);

ogAddImageLayer(globe, 'url: ["http://myglobe.com/img/"], service: "osm"');

ogAddElevationLayer(globe, 'url: ["http://myglobe.com/elv/"], layer: "elv"');
```



OpenWebGlobe Objects



A simple «hello world» with OSM Data (JavaScript)

```
<script type="text/javascript" src="openwebglobe.js"></script>
<script type="text/javascript">

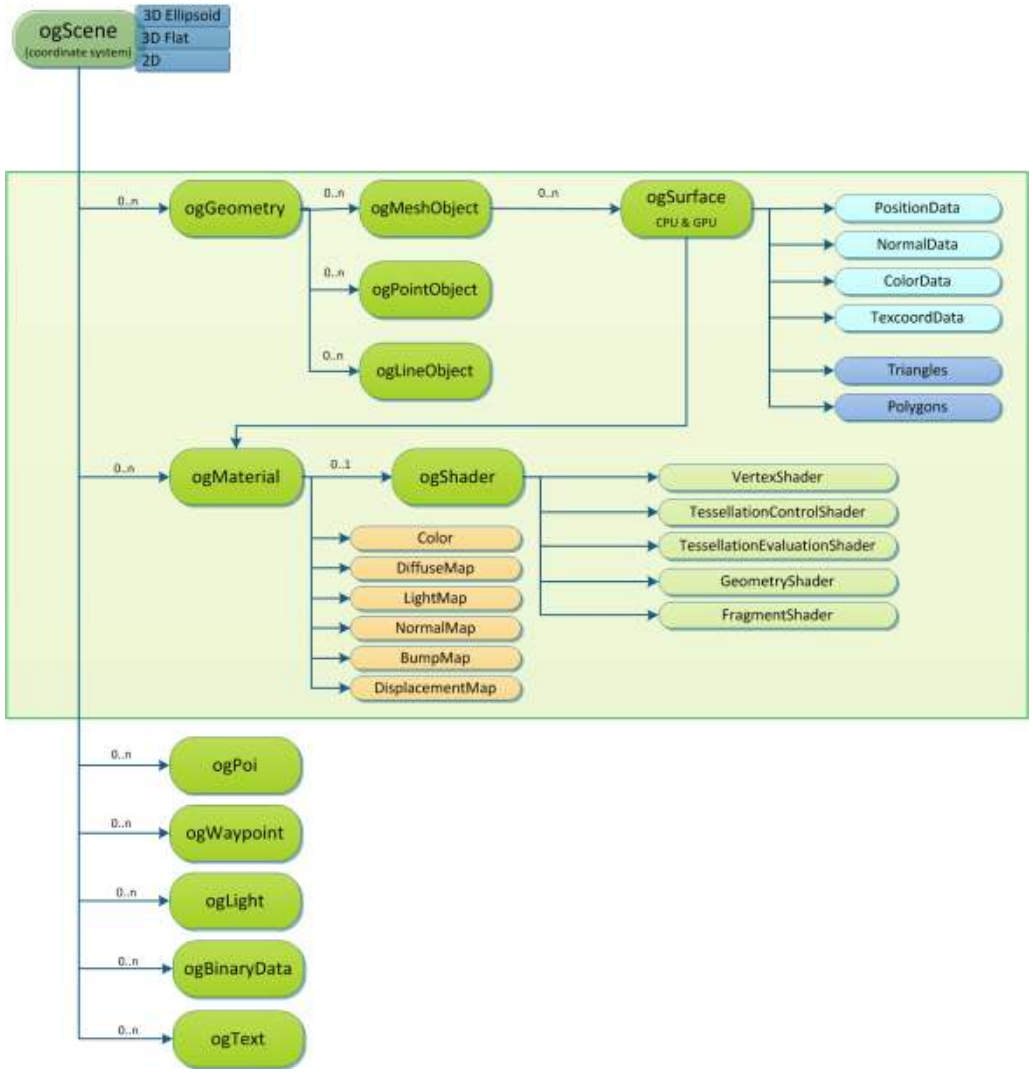
function main()
{
   var ctx = ogCreateContextFromCanvas("canvas", true);
   var globe = ogCreateGlobe(ctx);
   var imgOpenStreetMap =
   {
      url     : ["http://a.tile.openstreetmap.org",
                 "http://b.tile.openstreetmap.org",
                 "http://c.tile.openstreetmap.org"],
      service : "osm"
   };
   ogAddImageLayer(globe, imgOpenStreetMap);
}
</script>
<body onload="main()">
<div style="text-align: center">
<canvas id="canvas"></canvas></div></body>
```

A simple «hello world» with OSM Data: Adding POI




```
poilayer =
ogCreatePOILayer(globe,"airports",customTextstyle,customIconstyle);
var Poi =
   {
         "icon"     : "http://.../airport.png",
         "text"     :  "ZRH - Zürich",
         "position" :  [8.55, 47.4525, 1000]
   };
ogCreatePOI(poilayer,Poi)
```

More OpenWebGlobe SDK Objects



Data Processing

Storage: **-120 Terabyte**
-1.4 GB/s Sequencial Read
-850 MB/s Sequencial Write

Headnode: **-**
(Management of Compute Nodes)

Compute Nodes: **- Data Processing**



Data Processing Workflow

Create a Layer
Add Data to layer
Triangulate Data *(only if elevation data)*
Resample / Build all LOD

Update
Deploy

All Data Processing code is OpenSource
(MIT-License)

How we use OpenStreetMap Data

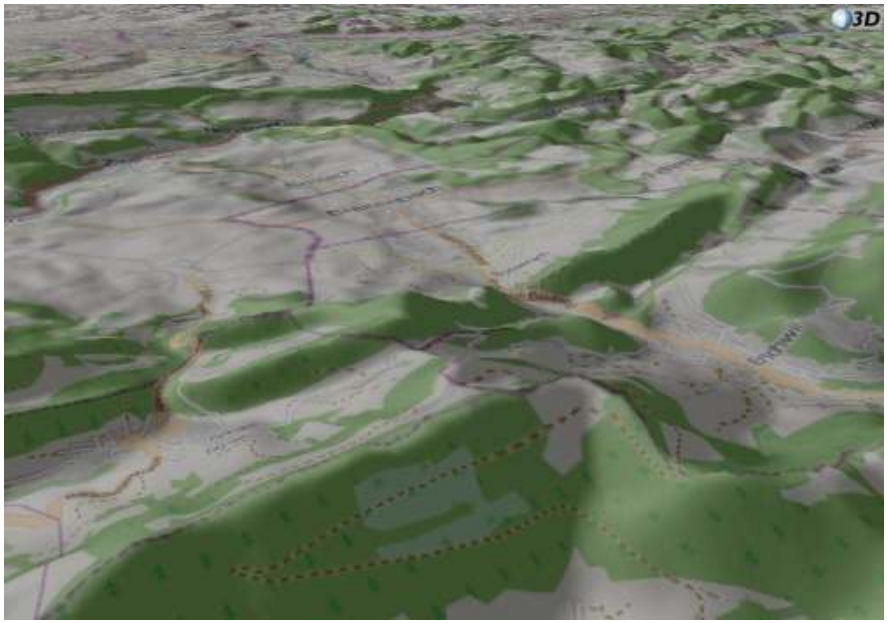






- **We render custom tiles: the «background» is transparent**
- **With transparent tiles we can combine different datasets**
- **We can combine it with orthophotos or by using hillshading tiles**

Combining OSM with hillshading and orthophotos





OpenWebGlobe on GitHub

- **The code is available on Github. OpenWebGlobe uses the «Closure Compiler» for JavaScript.**

- **A precompiled «SDK» is available together with some tutorials:**
  - **Text** (just draw some text)
  - **Texture** (just blit a texture)
  - **Globe** (display a simple globe)
  - **OpenStreetMap** (display globe with OSM data)
  - **Picking** (Click on Terrain to retrieve position including elevation)
  - **POI** (some examples with POI and custom styles)
  - **3D Geometry** (some examples of importing 3D-objects)
  - **Camera** (update position, orientation, have several cameras in a scene)
  - **Canvas as a billboard** (use HTML5 canvas as texture including video)

Coming soon: Point Cloud streaming support



Thank you!



OpenWebGlobe



OpenWebGlobe



martin.christen@gmail.com